

## Session 1:

# Overview of Electronic and Algorithmic Trading

Robert Kissell, Ph.D.

[Robert.Kissell@KissellResearch.com](mailto:Robert.Kissell@KissellResearch.com)

Kissell Research Group  
[www.KissellResearch.com](http://www.KissellResearch.com)

# Session I: Outline



This Session: Overview of Electronic and Algorithmic Trading

- Financial Markets Landscape
- Trading Algorithms: VWAP/WAP, POV, Arrival Price, Front- and Back-Load, Liquidity Seeking, Optimal
- Trading Venues: Exchanges, ATS/ECNs/Dark Pools
- Broker Compensation: Agency or Principal Bid (Capital Commitment)
- Market Order Book & Order Types
- Smart Order Routing
- Venue Revenues: Commission-Based, Maker-Taker, Inverted Maker-Taker

Next Session: Transaction Cost Analysis (TCA)

- Python Examples
- Post-Trade Analysis
  - Calculations behind Trading Algorithms
  - Implementation Shortfall & Trading Costs
  - Evaluate Brokers and Algorithmic Trading Performance (ex-post/historically)
- Pre-Trade Analysis
  - Estimating Costs, Evaluate Trading Strategies, Optimize Algorithms (ex-ante/forecast/predictions)
  - Estimate and build Market Impact Model

# Electronic and Algorithmic Trading

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# What is Electronic Trading?



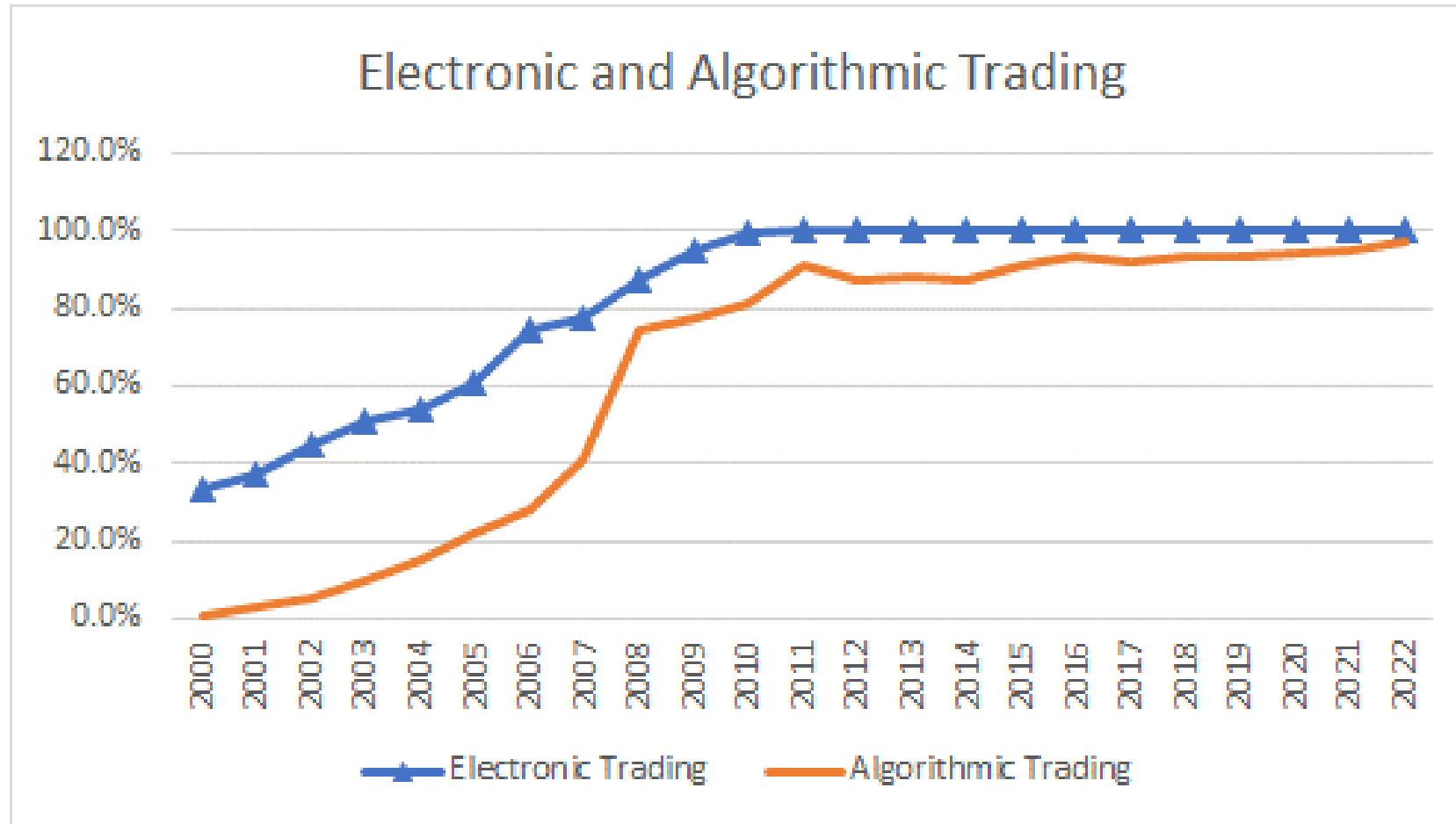
- Electronic trading is the process of entering and transacting order over a computer network.
- Historically, orders were sent to brokers via phone and/or fax and were executed via human traders and manual intervention. Traders were tasked with achieving the best price for their clients and would utilize the services of specialists and market makers.
- But now,
  - Markets are completely electronic, and all trading is preformed via computers.
  - Most market trades occur via trading algorithms that are making execution decisions in real-time based on actual market conditions and stocks trends.
- Most importantly, trading algorithms are making decisions without human intervention.

# What is Algorithmic Trading?



- Algorithmic trading is the computerized execution of financial instruments following a set of pre-specified trading rules and instructions.
- These instructions specify how fast or slow to transact the order in the market, and at what prices, and trading venues.
- Traditionally, investors would send their order to a broker who would then transact the order in the market using best efforts and their expertise.
- But with the changing landscape, human traders are no longer best equipped to handle the large quantity of financial data manually. Instead, human traders rely on sophisticated computer algorithms to execute orders based changing market conditions, quotes, and trends.
- Traders need to be proactive in the Algorithmic Trading environment and are required to specify trading goals based on the underlying investment objective of the fund.

# Electronic and Algorithmic Trading



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# Why are Algos so Important?

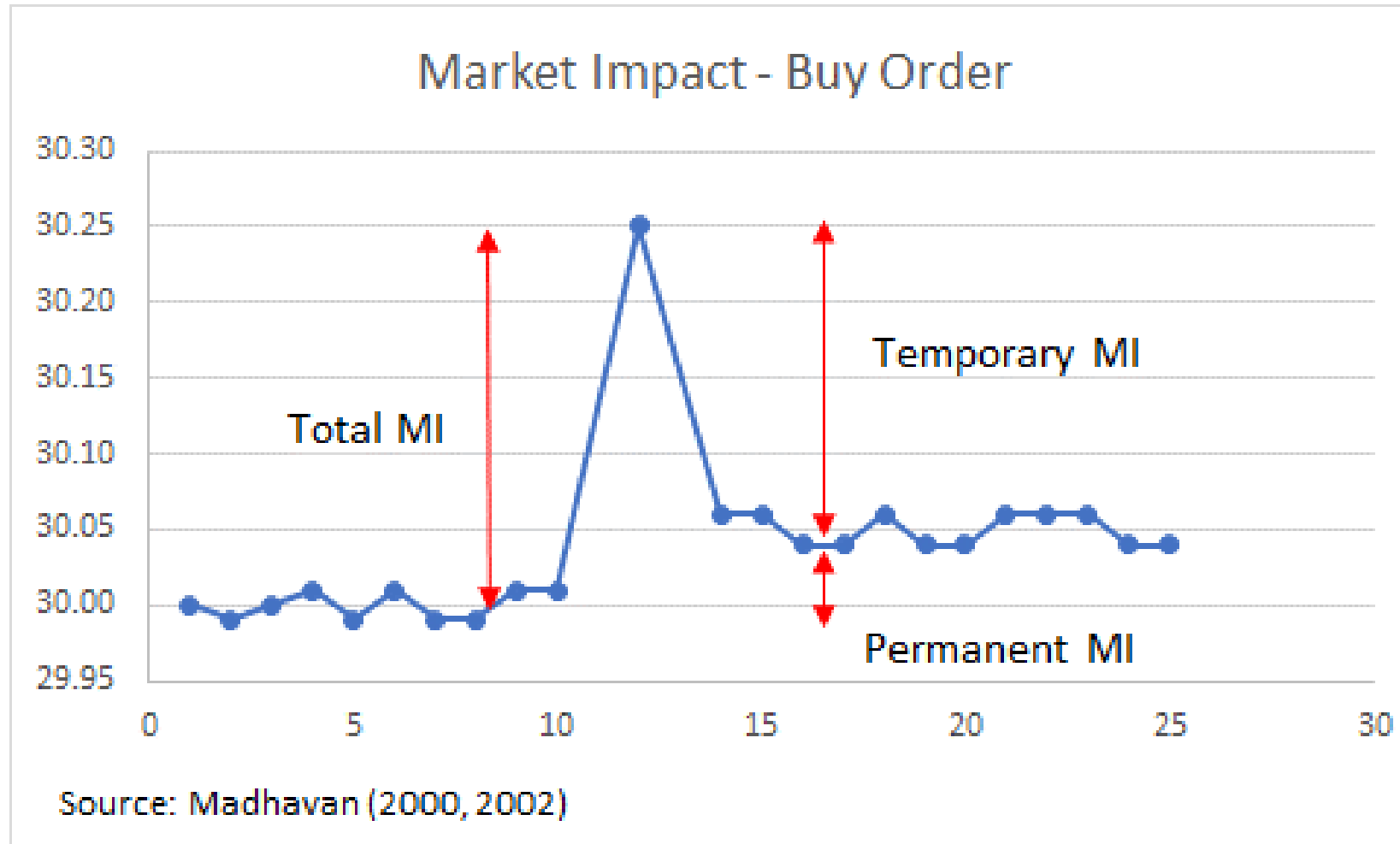


- Before we get started, it is important to say that Institutional Orders are usually very large, e.g., much more than 100 shares.
  - These large orders will cause price impact in the stock you are transacting.
  - For example, these buy orders cause the market price to increase and sell orders cause the market prices to decrease.
- How do investors combat this phenomena?
  - Slice the order and trade over time.
  - Balance market impact and timing risk.
- Select a trading strategy that is consistent with the investment objective.

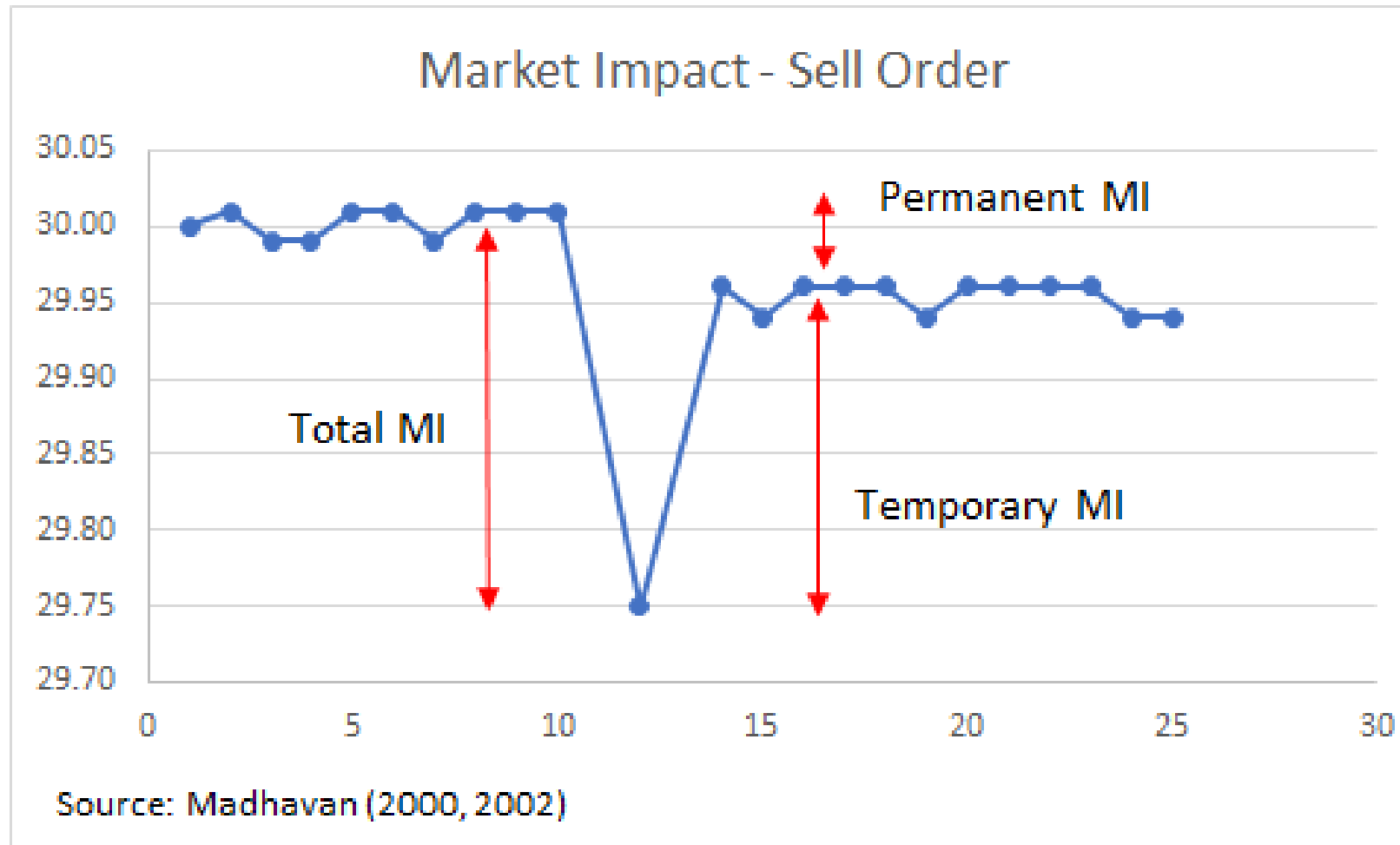
# Market Impact / Price Impact

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# Market Impact Cost – Buy Order

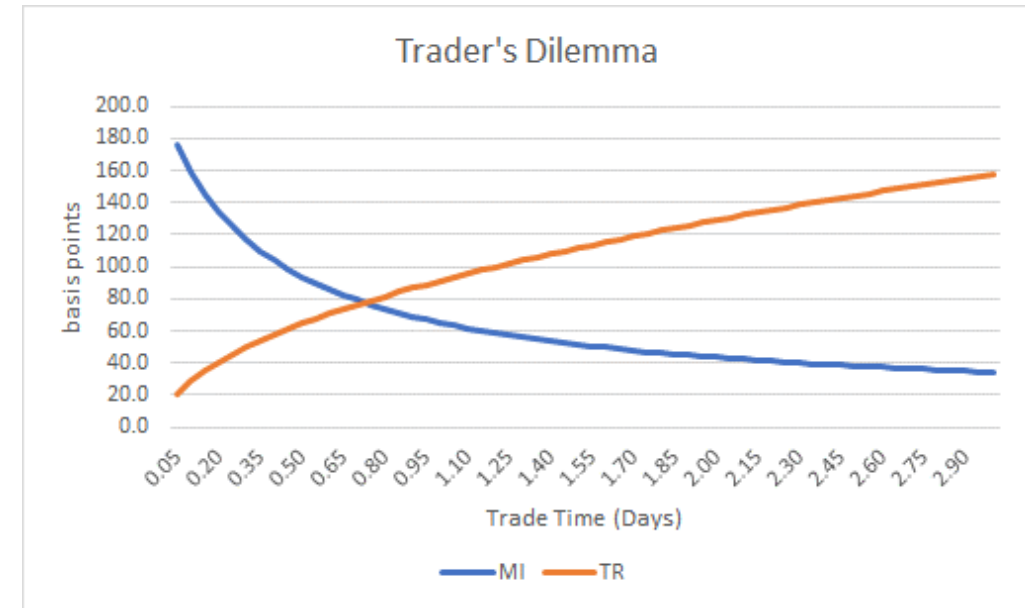


# Market Impact – Sell Order



# Trader's Dilemma

- Market Impact (MI) and Timing Risk (TR) are conflicting expressions.
- Market Impact is a decreasing function with trade time. Trading faster incurs higher MI and trading slower incurs less MI.
- Timing Risk is an increasing function with trade time. Trading faster incurs lower TR and trading slower incurs higher TR.
- Trader's Dilemma:
- Trading too fast will incur too much market impact cost but trading too slow will incur the order to too much market exposure risk.
- In other words, trade too fast and you move the market but trade too slow and the market moves you.



- Investors are tasked with determining the appropriate algorithm and execution strategy that will balance the tradeoff between Market Impact and Timing Risk.
- This is based on order characteristics, risk-aversion, and motivation of the investment decision.

# Solving Trader's Dilemma

## Solving Trader's Dilemma:

- Investors solve the Trader's Dilemma via optimization.

$$\text{Min } MI + \lambda \cdot TR$$

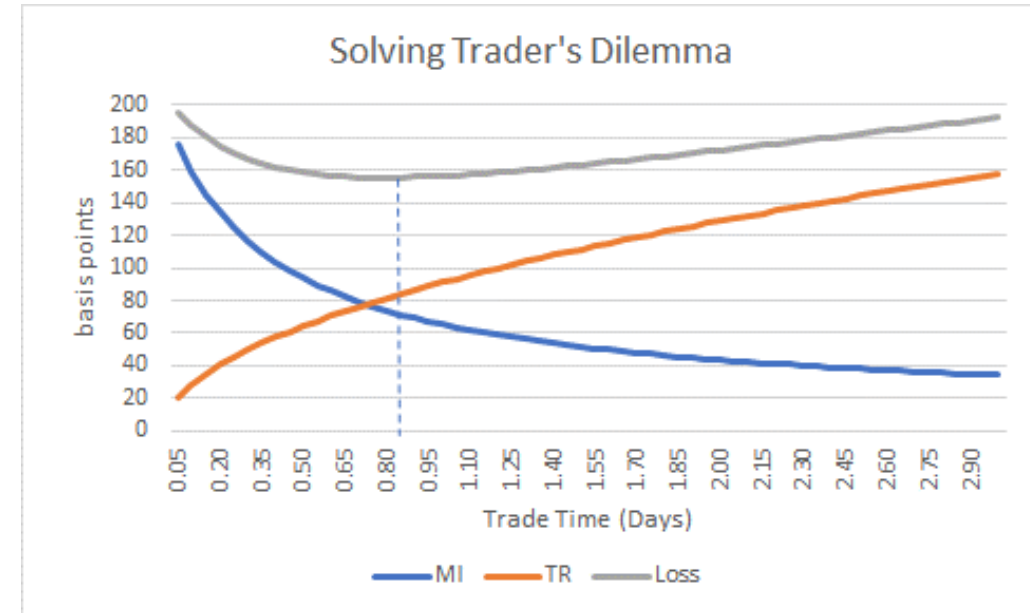
where,

MI = Market Impact Cost

TR = Timing Risk (uncertainty)

$\lambda$  = Investor's level of risk aversion

- The solution to the optimization problem will be the optimal time to complete the trade given market impact cost and timing risk, and the investor's level of risk aversion.

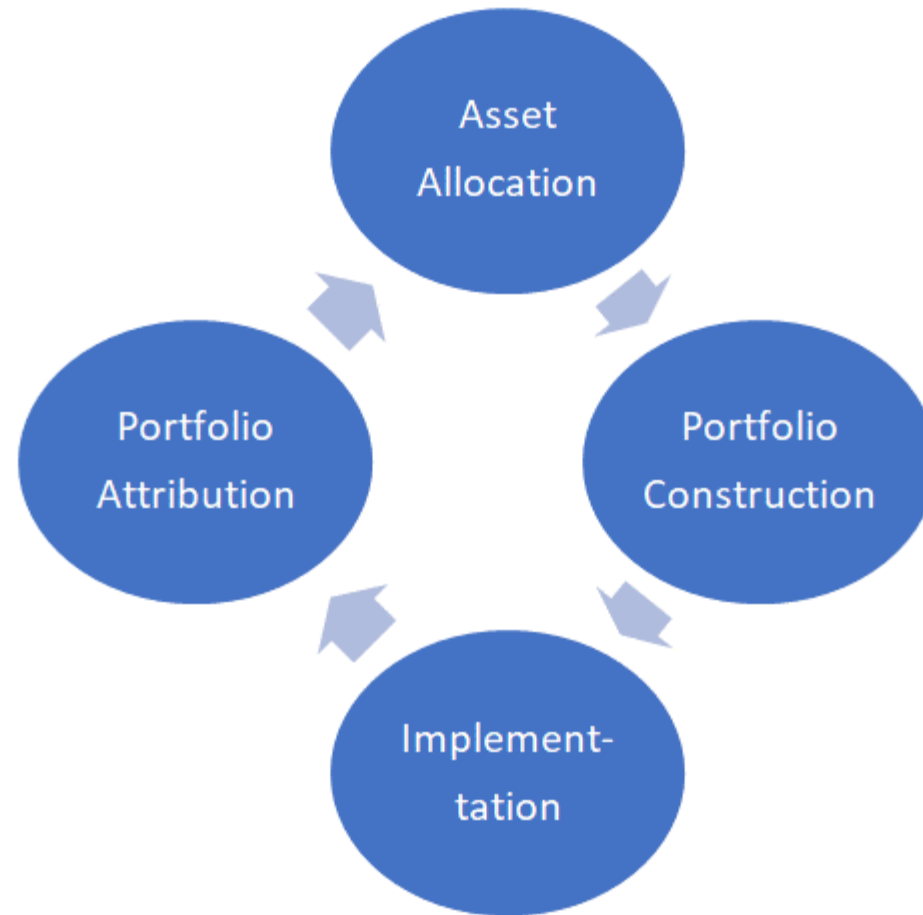


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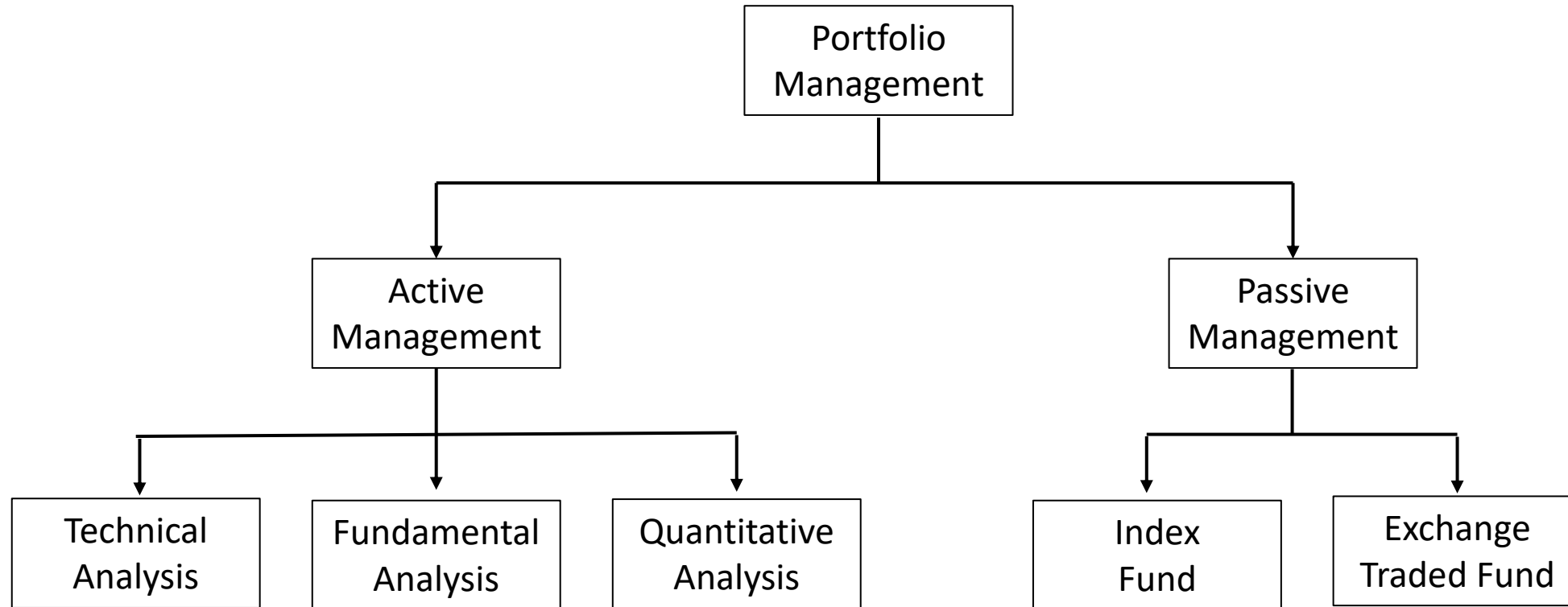
# Financial Markets Landscape

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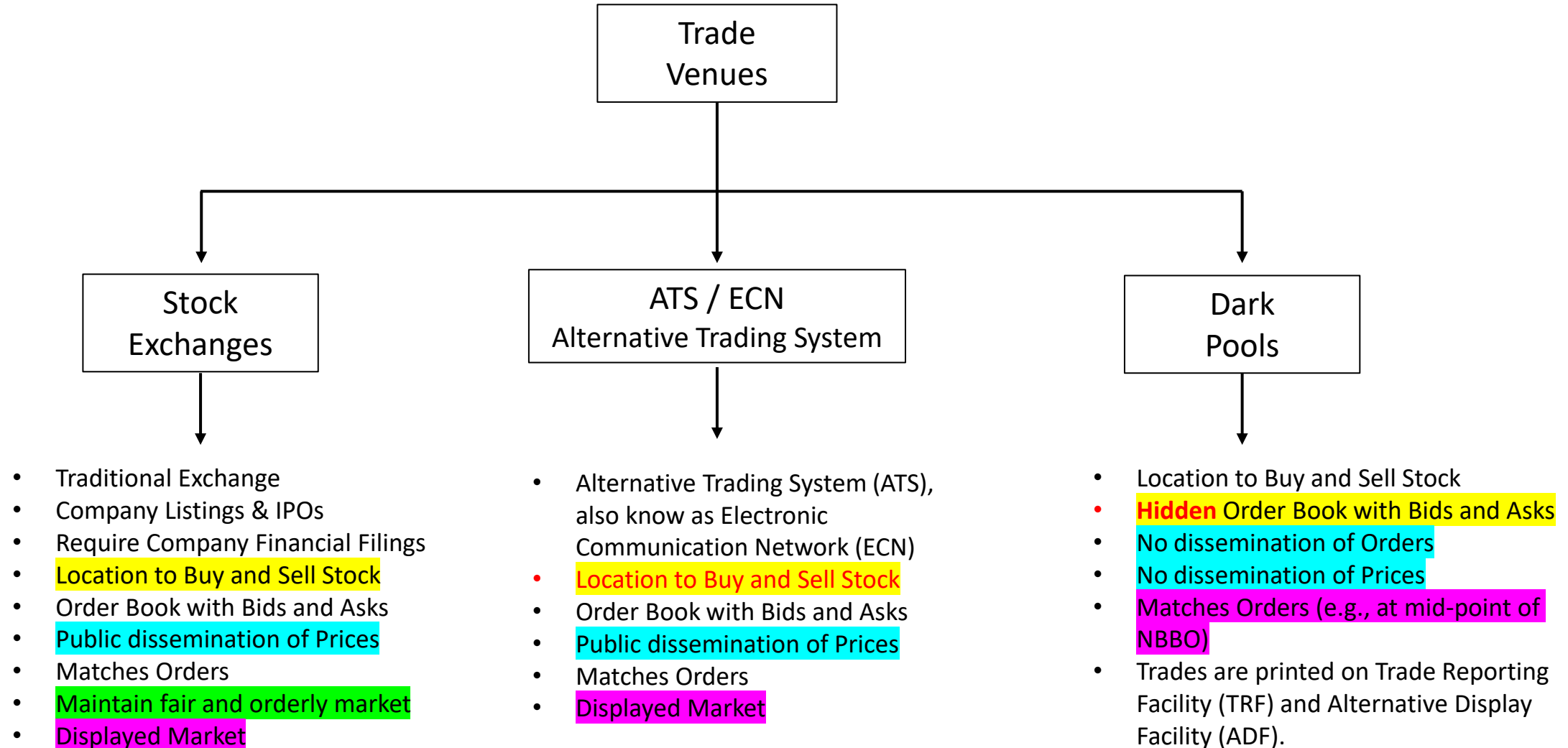
# Investment Cycle



# Portfolio Construction



# Electronic Trading Environment: Trade Destinations



# Electronic Trading Environment: Trade Destinations



**Stock Exchange:** Traditional Exchange where companies list their stock for execution. However, the specialist function has been replaced by a Designated Market Making (DMM) function and the Market-Maker has been replaced by Qualified Market Makers (QMM). An exchange display bids and asks to the public and provide incentives through rebates for providing liquidity to their venue. These new Auto-Market Makers (AMM) are required to maintain quotes for a specified amount of time (as define by the exchanges) in order to qualify for the rebates. Exchanges publicly display their best bid and best ask prices to the market, and clients can view the full depth of book.

**Alternative Trading System (ECN/ATS):** A destination where stocks trade like an exchange, however, these venues do not list companies or provide company about the company. Venues originated the rebate revenue model to attract order flow. Venues publicly display their best bid and ask prices to the market. Clients can view the full depth of book of the venue.

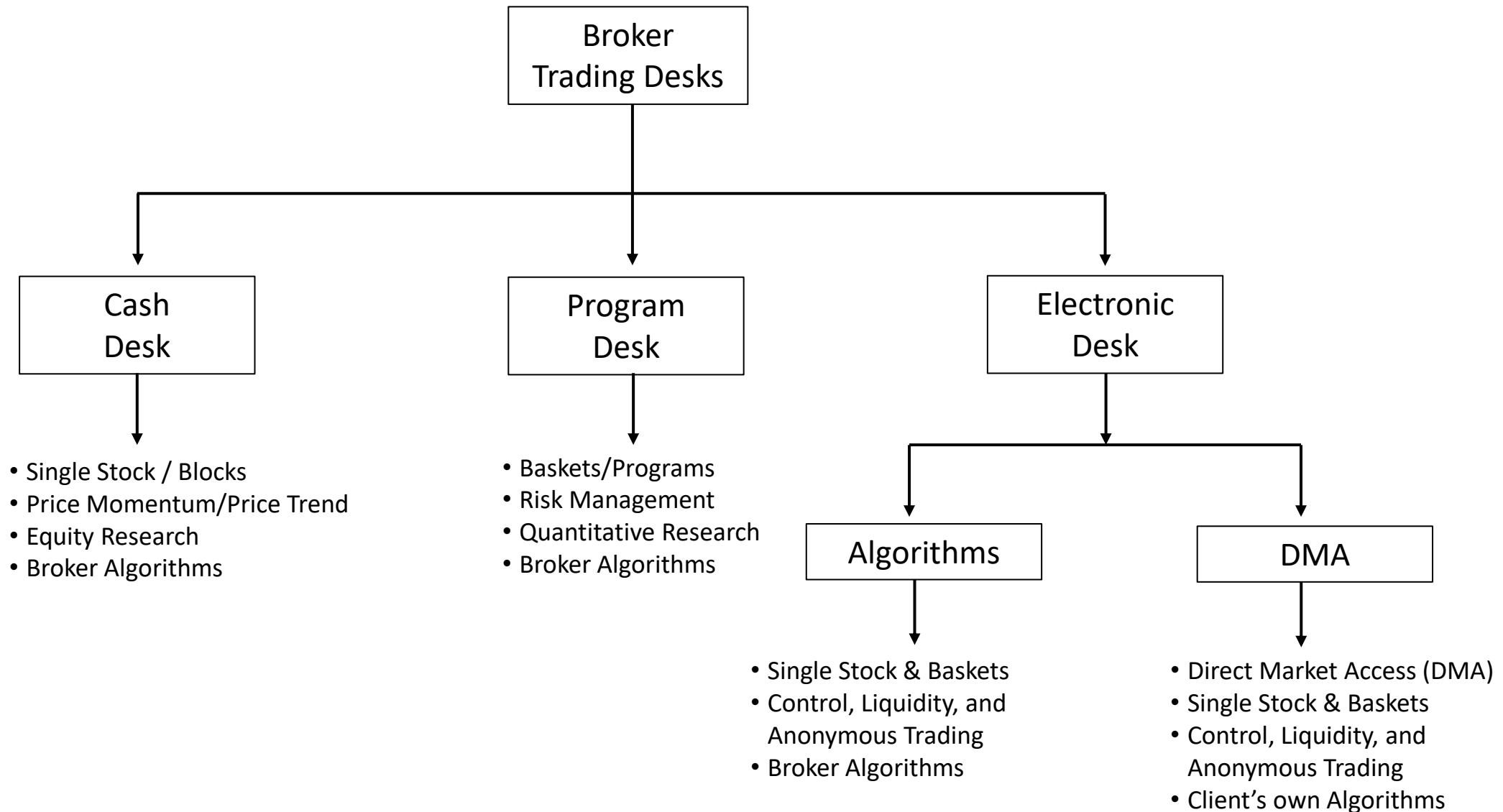
**Dark Pool (Crossing Network):** a dark pool or crossing network is a trading destination that does not publicly display any of their orders or quotes to the public. Their order book is complete hidden to the marketplace. However, due to competition and market regulation, dark pools need to trade at the best available market prices displayed to the public from exchanges or venues or better. The advantage a dark pool is that the customer will often trade within the bid-ask spread and will not disseminate any order information to the marketplace through their transaction.

# Types Dark Pools



- Broker-Dealer Dark Pools
- Independent Dark Pools
- Market Making Dark Pools

# Implementation: Where do Investors Trade



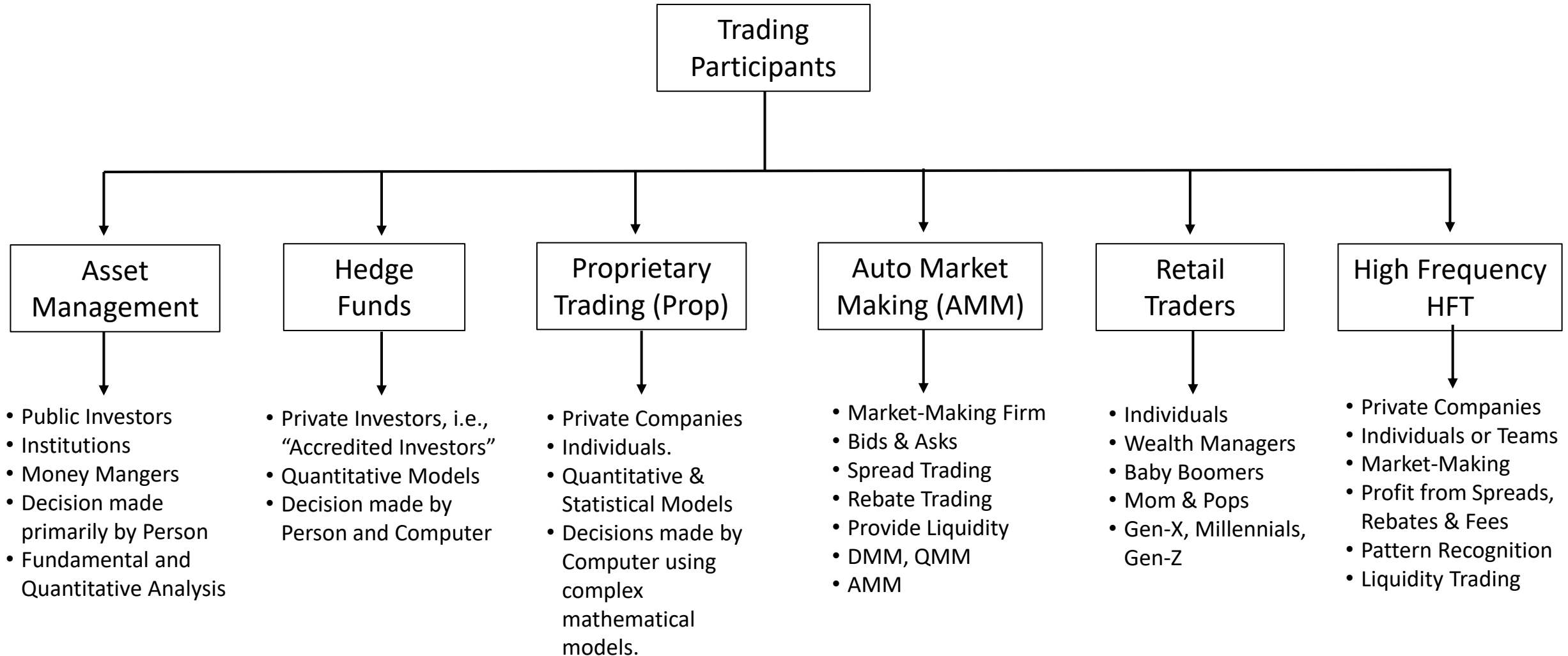
# Broker-Dealer Trading Desks

| Function                    | Cash Desk                                | Program Desk                                | Electronic Desk                  |                                    |
|-----------------------------|------------------------------------------|---------------------------------------------|----------------------------------|------------------------------------|
|                             |                                          |                                             | Algorithmic Trading              | DMA                                |
| Trading:                    | Single Stock / Blocks                    | Programs / Baskets<br>Index / ETFs          | Single Stocks / Baskets          | Single Stocks / Baskets            |
| Trading Concerns:           | Price Movement<br>Alpha                  | Risk Management<br>Cash Balancing           | Control / Anonymity<br>Liquidity | Control / Anonymity<br>Liquidity   |
| Primary<br>Research Product | Equity Research                          | Quantitative Research                       | Microstructure Research          | Microstructure Research            |
| Trade Execution:            | Broker Algorithms                        | Broker Algorithms                           | Broker Algorithms                | Investor Algorithms                |
| Decision Maker:             | Cash Trader<br>(w/Investor Instructions) | Program Trader<br>(w/Investor Instructions) | Investor<br>(Client)             | Investor<br>(Client)               |
| Sales Team:                 | Equity Sales<br>Corporate Access         | PT Sales<br>Quant Sales                     | Electronic Sales                 | Electronic Sales                   |
| Commissions                 | High<br>1-3 cents/share                  | Medium<br>1 cent/share                      | Low<br>0.5 - 1 cent per share    | Very Low<br>< 0.35 cents per share |

# Market Participants

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# Who are the Trading Participants



# Market Participant Type

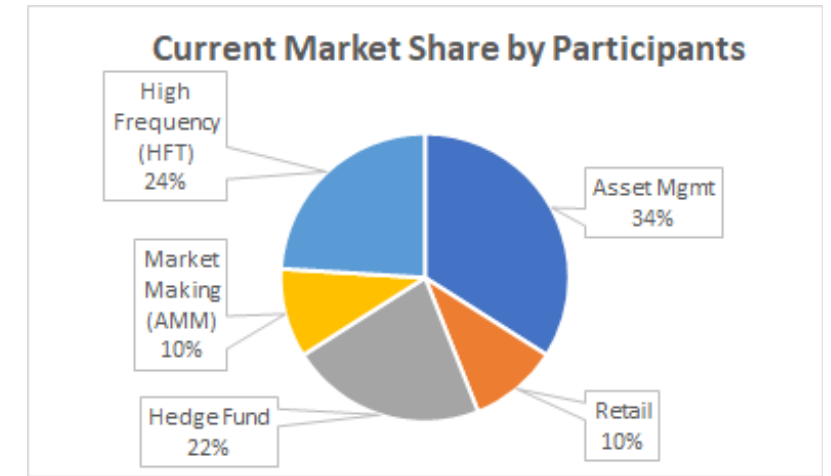
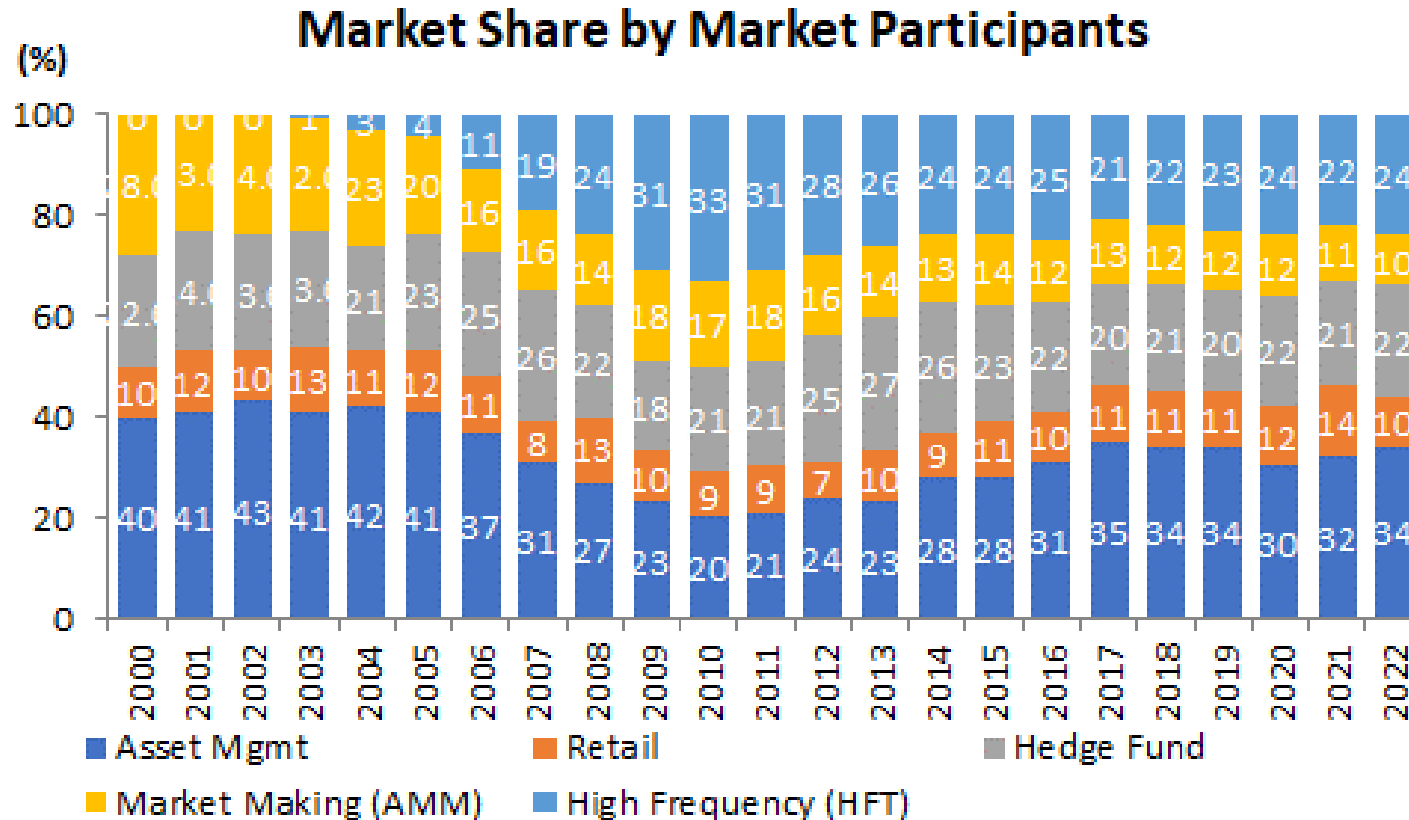


Brokers = Investment Banks, Full-Service, Bulge-Bracket, Research Brokers, etc. These are the market participants who facilitate the trade.

## Participants

- Asset Management = Institutions, Mutual Fund, Index Fund, Insurance Companies
- Hedge Funds = Quant & Non-Quant
- Market Makers = Liquidity Providers, Specialist / Market Makers, NYSE-DMM, Nasdaq-QMM, AMM-Brokers
- Retail Investors = Individuals, Wealth Managers, Millennials, Gen-Z
- Proprietary Trading Firms (Prop) = Prop Firms. Categorized as Hedge Funds, Liquidity Providers, and/or HFT Firms.
- High Frequency Trader (HFT)

# Market Participants



Source: Kissell Research Group (2022)

# Trading Algorithms

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# Types of Algorithms



- **Execution-Algo** = Algorithms are tasked with transacting an order that was entered by a portfolio manager and determined outside of the system. Investors specify the transaction rules to the algorithm at beginning of trading.
- **Quant-Trading** = Algorithms that are tasked with making the investment decision and then transacting the order in the market to earn a profit. These algos are also often referred to as “Profit-Seeking” algos. The investment decision is often based on a quantitative approach such as portfolio optimization, mathematical model, or stat-arb approach. Quantitative algorithms make investment decisions based on a mathematical model and seek to earn an investment profit. Once an investment opportunity is uncovered in the market the algorithm will immediately execute that decision in a manner appropriate for the investment opportunity. These algorithms seek to profit from their investments and are used by quantitative traders and proprietary trading firms.
- **High Frequency Trading (HFT)** = Algorithms that provide liquidity to the market though entering bids (buy orders) and asks (sell orders) to trade venues and seek to earn a “trading profit” (e.g., profiting through short-term buying and selling of stocks based on investor demand. These “HFT” algorithms will earn a profit via three means: i) profit from the bid-ask spread by buying at the bid (lower price) and selling at the ask (higher price), ii) via market rebates where trade venues provide a rebate (monetary incentive or reward) to investors who provide liquidity to that venue, and iii) by uncovering buying and/or selling patterns in the stock and using that information to their advantage.

# High Frequency Trading (HFT)



- Liquidity Provider / Liquidity Trading
- Proprietary Trading
- Auto Market-Making (AMM)
- Spread Trading – Bids and Asks
- Fees & Rebates
- Seek to uncovering Investor Trading Patterns - using publicly available market data (prices and quotes) and information.
- Use of all trading venues including Exchanges, ATS, and Dark Pools

- **Aggressive:** Get Me Done. Sweep all Liquidity available at My Price or Better. Uses more market orders during times of favorable prices to capture these better prices. More market orders than limit orders.
- **Working Orders:** Balance the tradeoff between cost and risk, manage placement of market and limit orders. VWAP/TWAP, POV/Volume Participation, Arrival Price/Implementation Shortfall, Optimal Strategies. **Focus of Session 2.**
- **Passive:** Usage of crossing systems and dark pools. Liquidity Seeking Algorithms. Will use more limit orders than market orders. More limit orders than market orders.

# Algorithm Slicing Strategies



- **Time-Based:** Trade a specified number of shares at specified times during the day. E.g., execute 25,000 of the order between 9:30am and 10:00am. The same number of shares will occur in these time periods regardless of market volumes and market prices.
- **Volume-Based:** Participate with a specified amount of volume. E.g., POV=10% indicates that your trades will constitute 10% of market volume. More trading will occur in times of higher market volume and less trading will occur in times of lower market volume.
- **Price-Based:** Traders may execute more or less aggressively based on market prices.

# Algorithm Updating



- **Static:** An algorithm that does not update, revise, change, or modify its trading strategy during the execution of the order. Once the parameters for these algorithms are specified, they can only be changed by the trader making a manual revision to the algorithm.
- **Dynamic:** An algorithm that revises its trading strategy and parameters during execution of the order based on changing and unexpected market conditions. These revisions are made by the algorithms itself and is often based on a specified mathematical equation or real-time optimization that considers market conditions, future expectations, and realized prices.

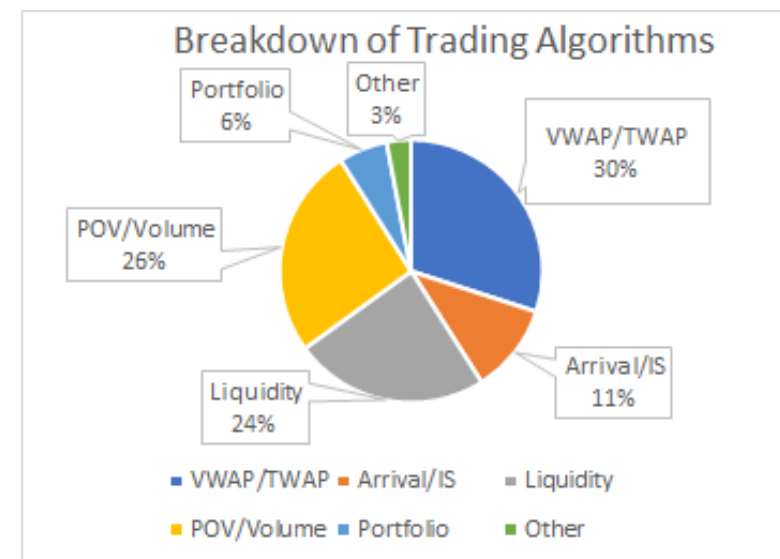
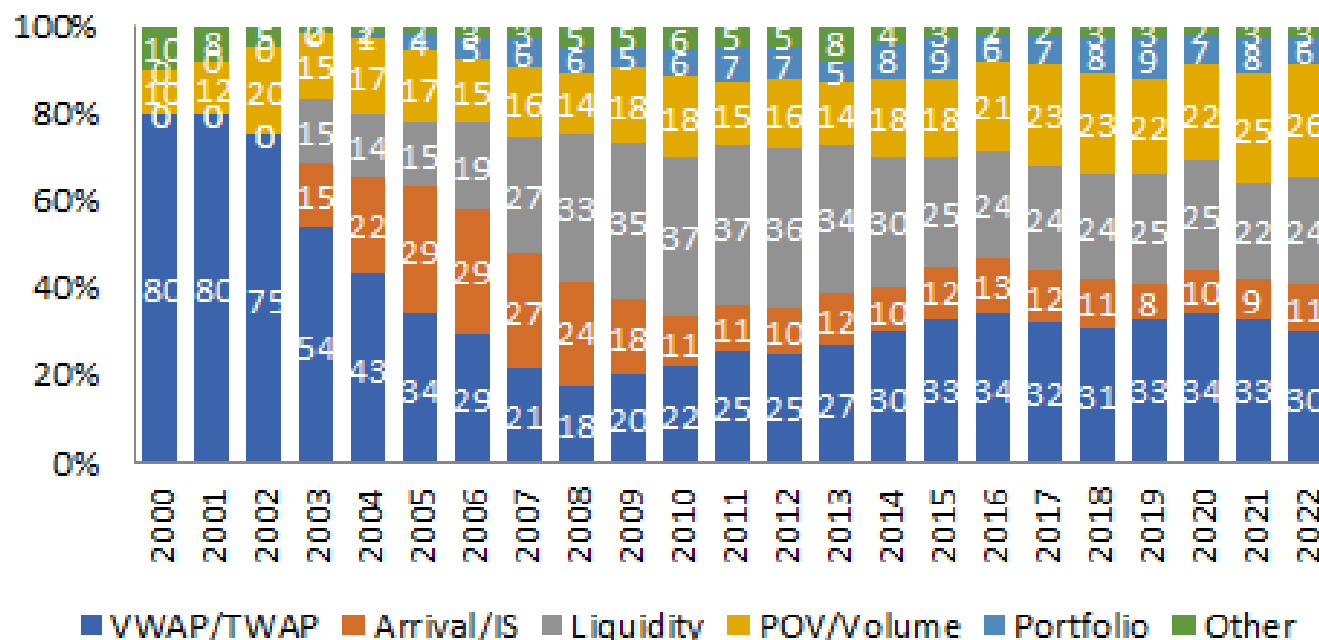
# Common Algorithm Types



- VWAP / TWAP
- POV / Volume
- Price Based: Arrival Price / Implementation Shortfall (IS)
  - Front-Load / Back Load
- Liquidity Seeking
- Optimal Strategy
  - Portfolio Algorithms

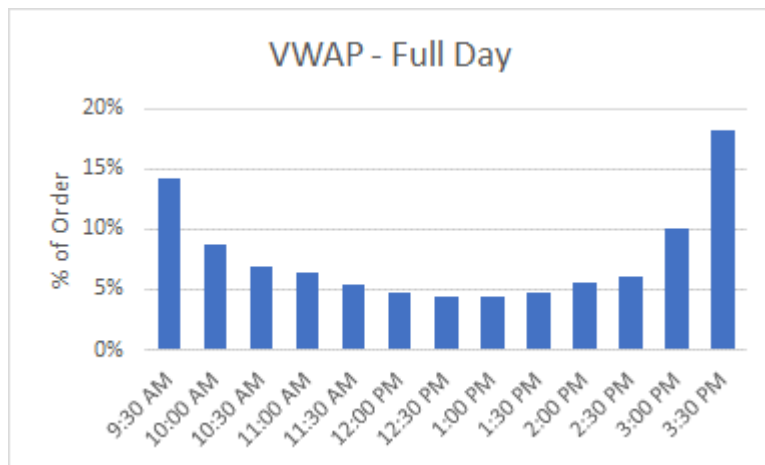
# Trading Algorithms

## Algorithmic Usage

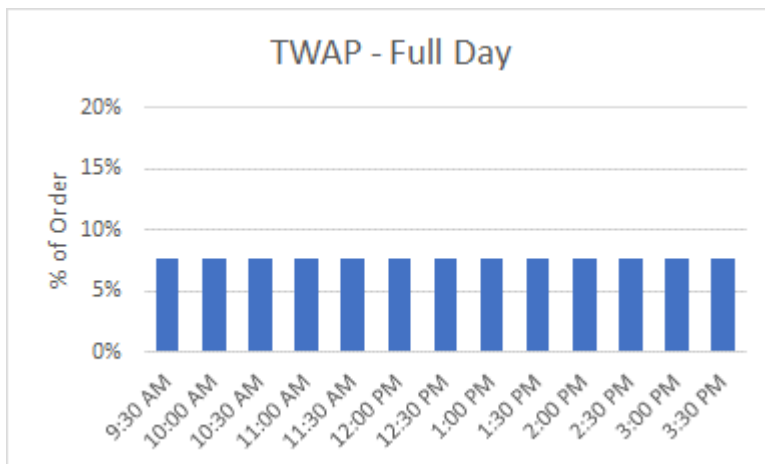


Source: Kissell Research Group (2022)

# VWAP/TWAP Algorithms



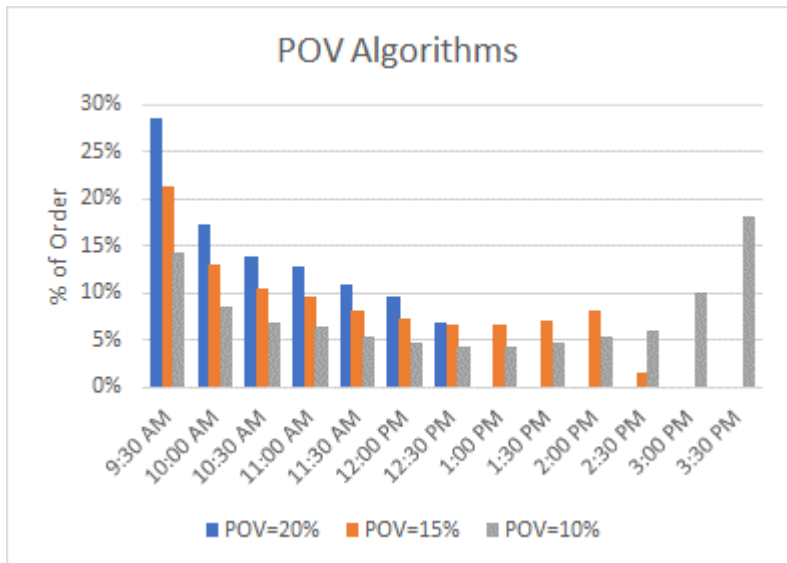
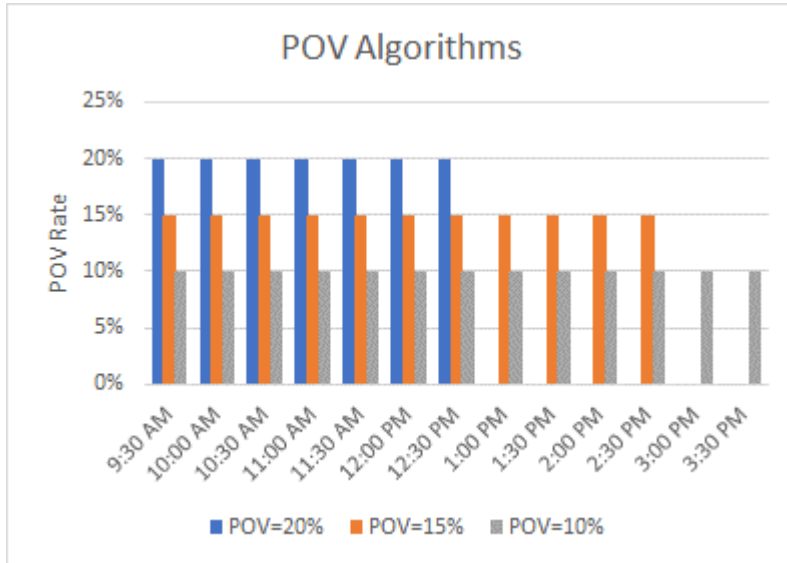
- A VWAP algorithm, e.g., Volume Weighted Average Price, is an algorithm that slices the order over the day or over a specified trading period based on the intraday volume profile of the stock. Historically, there has always been more volume traded at the open and close than during midday hours. More recently, there is even a greater amount of volume traded at the close than at the open.



- A TWAP algorithm, e.g., Time Weighted Average Price, is an algorithm that slices the order over the day or over a specified time period using equal share quantity amounts in each period. In statistics, this type of slicing is often described as a uniform distribution.

Source: Kissell Research Group (2022)

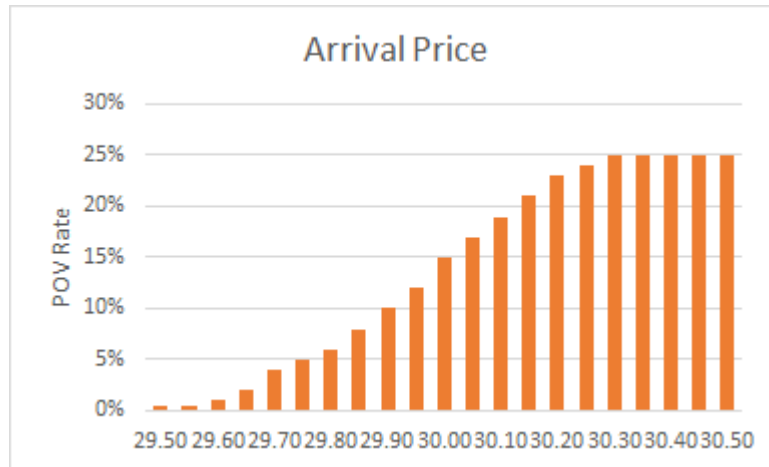
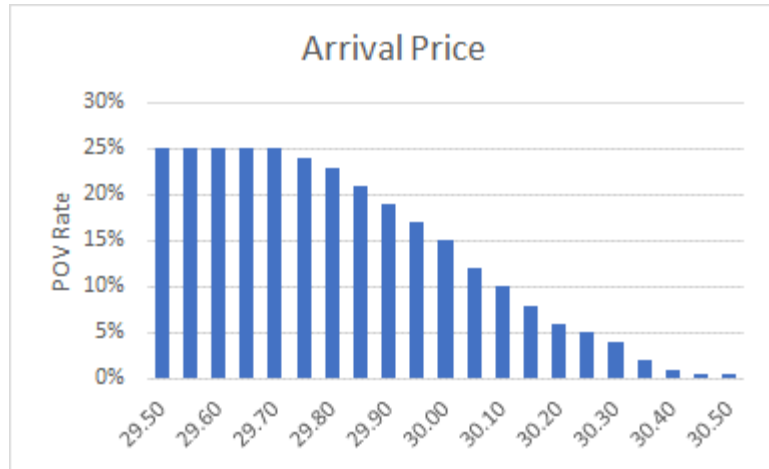
# Volume: POV Algorithms



- A POV algorithm, also known as a “Percentage-of-Volume,” “volume-participation,” “volume-follow,” or simple a “volume-based” algorithm is an algorithm that trades at a constant percentage of volume in each period until the order is completed.
- For example, POV=10% indicates that the algorithm is to participate with 10% of the market volume. This means that whenever 1000 shares trades in the market the investor will need to have transacted 100 shares.
- The graph on the upper left shows the POV rate (percentage of volume trading trade) for three different stocks. Orders with a higher POV rate will complete sooner than orders with a lower POV rate.
- The graph on the bottom left shows the % of the order that is expected to be transacted in each period. Notice how a constant POV results in different number of shares being transacted in each period. This is because stocks trade different amounts in each period during the day.

Source: Kissell Research Group (2022)

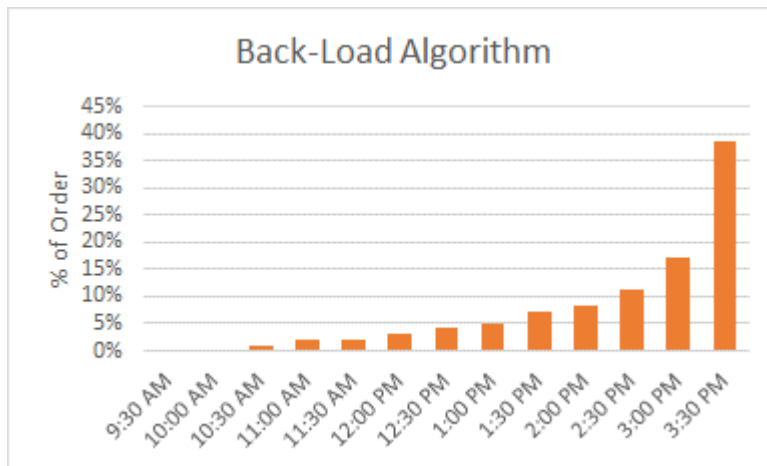
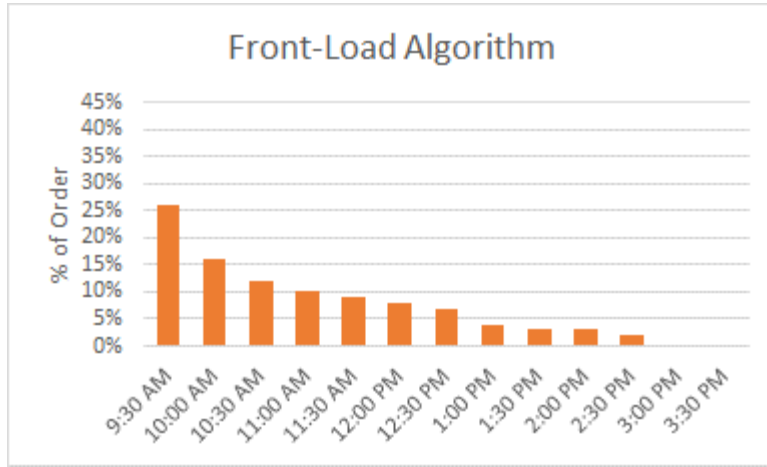
# Arrival Price Trading Algorithms



- An “Arrival Price” algorithm, also often referred to as an “Implementation Shortfall” algorithm is an algorithm that varies its trading rate based on market prices.
- An arrival price algorithm may trade aggressive (faster) or passive (slower) when market prices are at or near the arrival price of the order. As market prices move further away from the arrival price, either higher or lower, the arrival price algorithm will change its trading rate.
- It is important to mention that the trading behavior of Arrival Price algorithms are not consistent across brokers and vendors, and it is important for investors to understand exactly how the specific arrival price algorithm will transact shares in the market.
- Arrival price algorithms are often coupled with dynamic behavior.

Source: Kissell Research Group (2022)

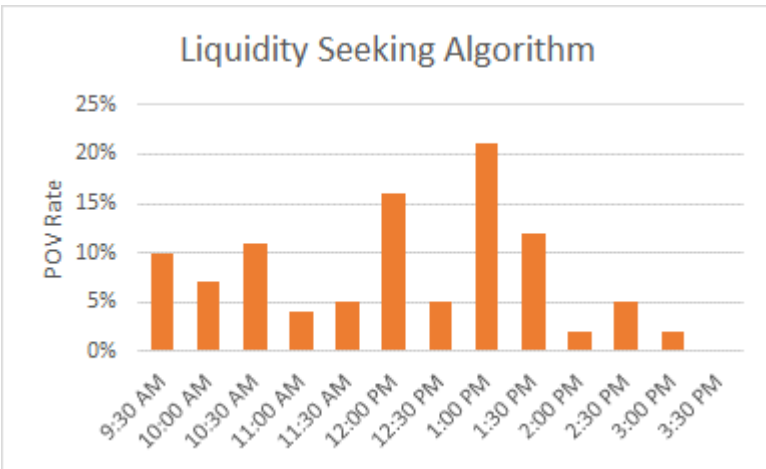
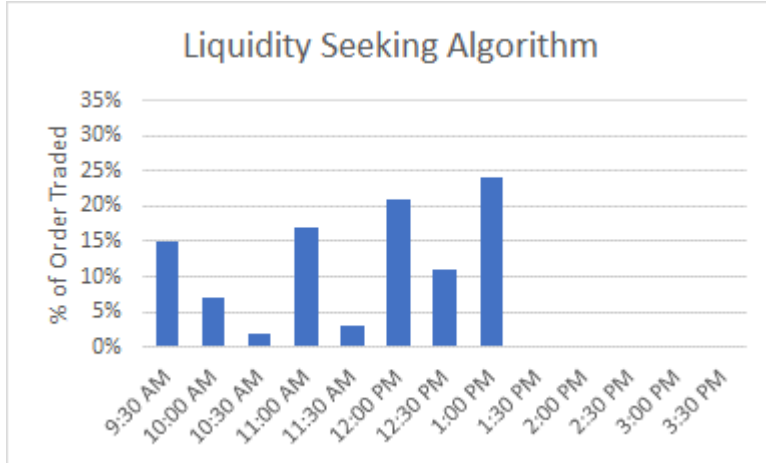
# Front-Load/Back-Load Trading Algorithms



- **Front-Load.** A front-load algorithm is time-slicing algorithm that trades more aggressive at the beginning of the order and more passive at the end of the order.
  - A front-load algorithm is often used by investors who are trying to capture prices as close to the arrival price as possible and still want to manage the trade-off between trading costs and timing risk.
- **Back-Load.** A back-load algorithm is a time-slicing algorithm that trades more passive at the beginning of the order and more aggressive at the end of the order.
  - A back-load algorithm is often used by investors who are trying to capture prices as close as possible to a future benchmark price, such as the close, and still want to manage the trade-off between trading cost and timing risk.

Source: Kissell Research Group (2022)

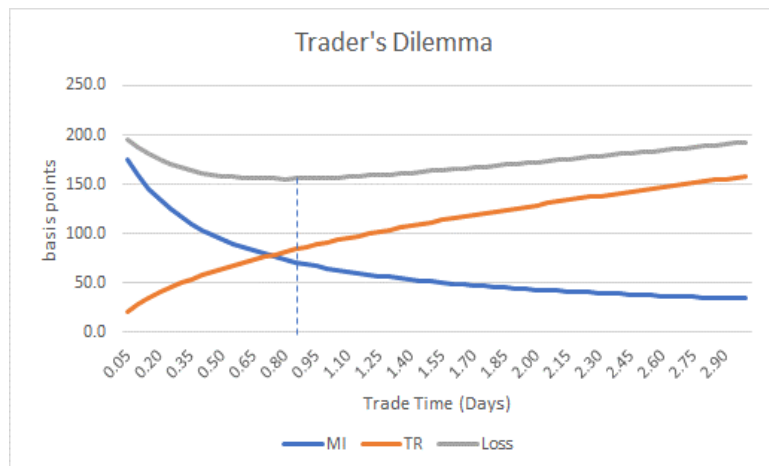
# Liquidity Seeking Algorithms



- **Liquidity Seeking.** A liquidity seeking algorithm is an algorithm that will utilize all trading venues (displayed and dark venues) to capture liquidity at a specified price or better. The liquidity seeking algorithm will often trade in an aggressive manner to capture as much liquidity as possible during times of favorable prices. The liquidity seeking algorithm will also make appropriate use of limit orders and dark pools.
- A liquidity seeking algorithm is also used by investors to execute shares at the tail end of the order. For example, trade aggressively to complete the order. In this situation, investors are not as concerned with information leakage if they can complete the order.
- Investors can also often specify how much to trade in a dark pool compared to the displayed venues.
- The graph in the upper left shows an example of a liquidity seeking algorithm that was able to transact at its specified price and complete the order by early afternoon.
- The graph in the lower left shows an example of a liquidity seeking algorithm that is not able to locate appropriate liquidity and requires a longer period to complete the order.

Source: Kissell Research Group (2022)

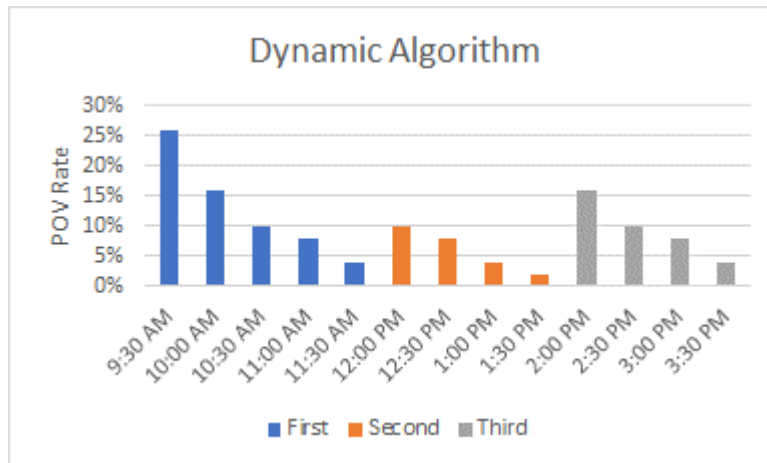
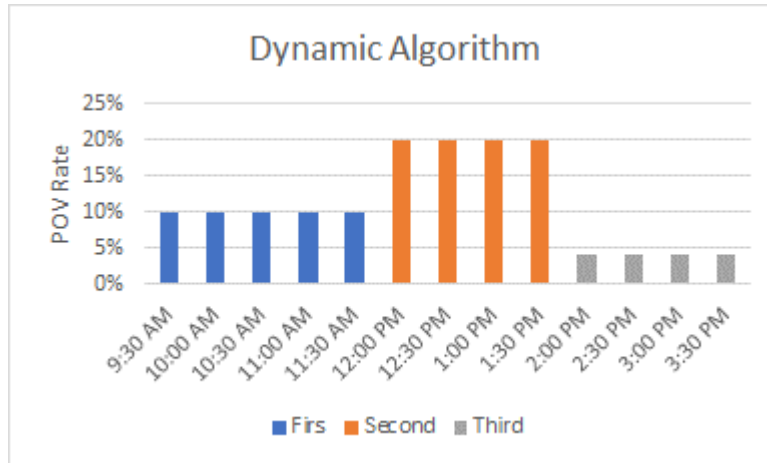
# Optimal Strategy Algorithm



- **Optimal Strategy.** An optimal strategy algorithm is an algorithm whose execution strategy (e.g., POV rate) is determined through an optimization process. For example, a common optimization specification is to solve the trader's dilemma where investors seek to balance the trade off between market impact and timing risk for a specified level of risk aversion.
- The graph in the upper left shows an example of how the optimizer will solve the trader's dilemma for a single stock. Notice how market impact cost decreases by trading over a longer period of time but timing risk (which is market exposure risk) increases when we trade slower. The optimizer will find the best solution for the investor based on their level of risk aversion.
- The graph in the lower left shows the actual trade schedule resulting from the optimization process and how many shares of the order to execute in each time period.

Source: Kissell Research Group (2022)

# Dynamic Algorithms



- **Dynamic.** A dynamic trading algorithm is an algorithm that adapts to changing market conditions such as prices and volumes based on investor specified trading rules and risk aversion specifications.
- These rules allow the algorithms to take advantage of market conditions and realized prices when appropriate.
- Algorithms can make often dramatic changes to the current trading rate. This is usually accomplished via a specified set of rules or equations, or via a re-optimization process.
- Dynamic trading logic is also often integrated into the algorithms mentioned above.
- The graph in the upper left shows the specified POV rate for a stock with dynamic optimization. In this example, the algorithm underwent three optimizations resulting in three different POV rates.
- The graph in the lower left shows the shares that will be executed in the market based on these optimal POV rates and actual market volumes and conditions.

Source: Kissell Research Group (2022)

# Execution Style

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# How Do Brokers Get Paid



Brokers get paid by executing order for investors.

- Agency Execution
- Principal Bid / Capital Commitments

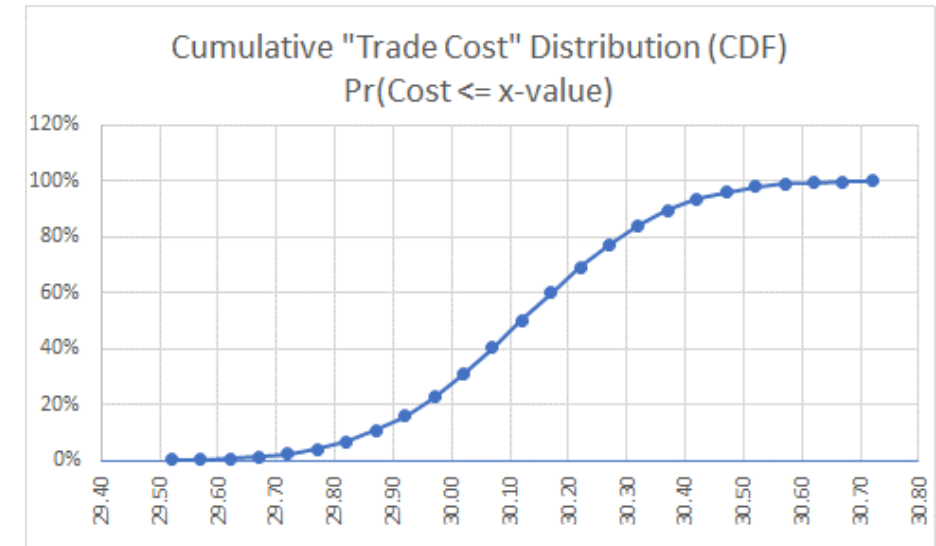
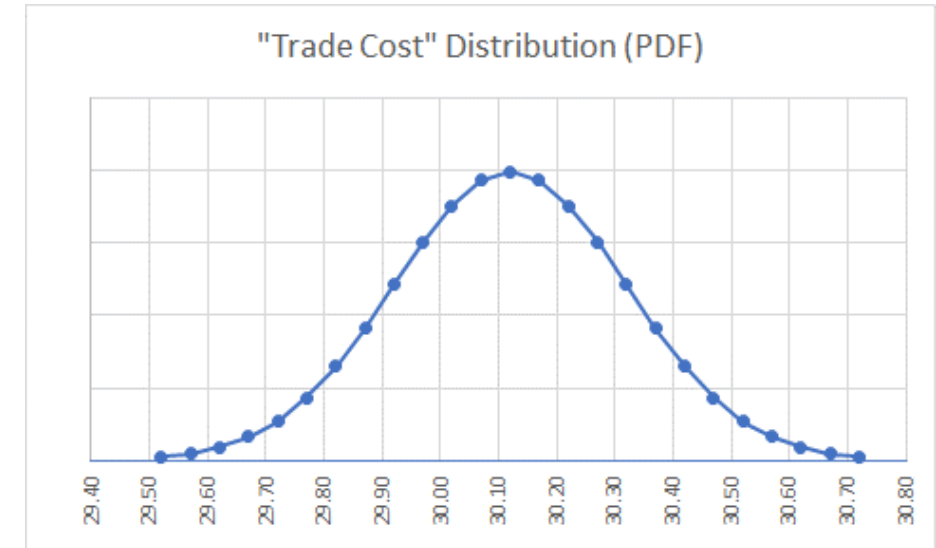
# Trading Types



- **Agency Execution:** the investor/fund pays the broker a “commission” to execute the order on a “best efforts” basis. Prices are unknown in advance and the investor incurs all market risk. If the prices increase a buy order will execute less favorably, and if it the prices decrease the buy order will execute more favorably. Vice versa for sell orders. In an agency transaction the broker profits the commission. Commissions are 1-3 cents per share.
- **Principal Bid / Capital Commitment:** the investor pays the broker a “premium” to execute the order. This premium amount is also known as a block bid, principal bid, or risk bid. In a principal bid, the investor receives a known or guaranteed price (e.g., current price, closing price, VWAP price) and transfers all risk to the broker. The principal broker receives the order increasing or decreasing their inventory position. If they can offset the acquired order (position) at a cost less than the bid amount they received they will profit, if not they will incur a loss. Brokers will always seek to hedge their risk in a principal bid transaction.
  - Principal Bid is much higher, say 5-25 cents per share.

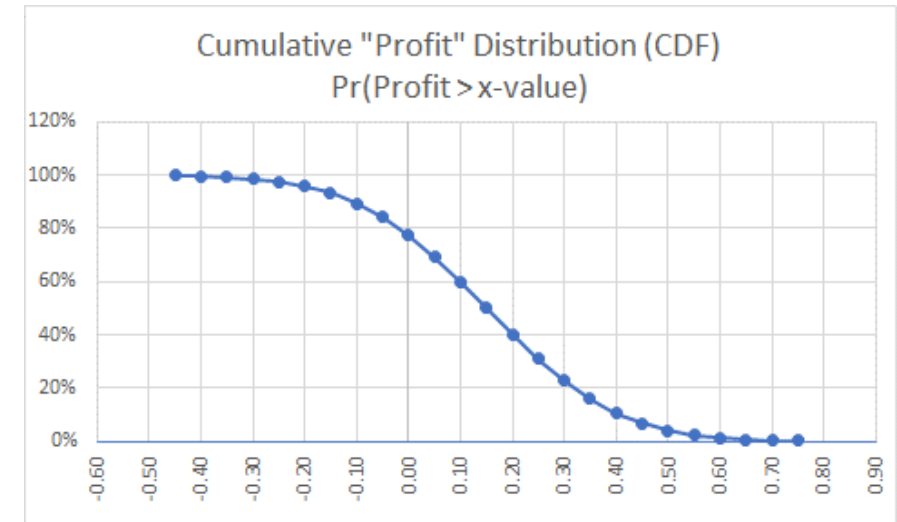
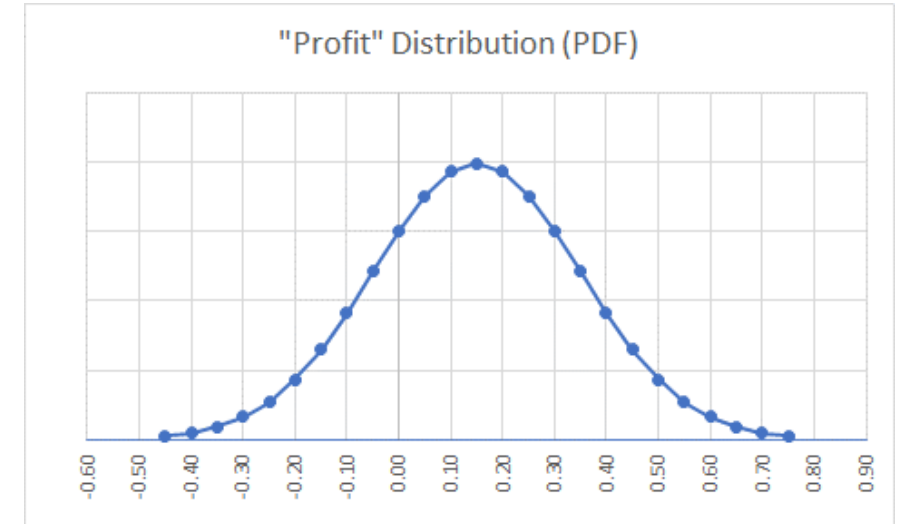
# Agency Execution: Example

- **Example:**
- An Investor wishes to buy 100,000 shares of stock RLK using an agency execution.
- The current price (arrival price) of RLK is Price=\$30.00. The market impact cost for the trade is MI=\$0.10 and timing risk is TR=\$0.20. The broker charges a commission of Comm=\$0.02/share.
- The expected execution price for RLK using an agency transaction including the commission is:
- My All in Price Price =  $\$30.00 + \$0.10 + \$0.02 = \$30.12$
- The standard deviation of the expected execution price is:
- TR = \$0.20.
- Therefore, the expected execution price is:
- $\$30.12 \pm \$0.20$ .
- The potential set of possible execution prices is illustrated in the Trade Cost PDF and CDF graphs.



# Principal Bid Transaction: Example

- **Example:**
- An Investor wishes to buy 100,000 shares of stock RLK using a principal bid transaction.
- The current price of RLK is Price=\$30.00. The broker charges the investor a principal bid of \$0.25/share for this transaction.
- In this situation, the investor receives an execution price of \$30.00 but pays the broker the bid amount of \$0.25. There is no risk for the investor. All risk is transferred to the broker. My all in Price \$30.25.
- The broker will need to purchase 100,000 shares in the market. The market impact cost for the trade is MI=\$0.10 and timing risk is TR=\$0.20.
- The expected profit for the broker is:
- Profit = Bid – MI Cost = \$0.25 - \$0.10 = \$0.15 with TR = \$0.20.
  - Expect Profit = \$0.15 +/- \$0.20.
- The broker has a 78% chance of earning a Profit.
- The broker's potential Profit distribution is illustrated in the Trade Cost PDF and CDF graphs.



# Market Order Book

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# Order Types



- **Market Order** = an order to immediately buy or sell stock at the best available market price.
- **Limit Order** = an order to buy or sell stock at a specified price or better.
  - **Bid Price** = a buy order. This is the price that an investor wishes to pay for stock. i.e., the buyer is bidding a specified amount to buy shares.
  - **Ask Price** = a sell order. This is the price that an investor wishes to sell shares. i.e., the seller is asking other investors to pay the specified value for shares. Also referred to as the Offer price. The price at which an investor is offering the stock.
- **Bid-Ask Spread** = the difference between the best bid and ask price. It represents the total cost of trading if an investor were to simultaneously buy and sell shares in the same stock. Also referred to as simply the Spread.
- **Order Book** = the collection of all bids and asks at a trade venue. Order Queue is most commonly Price-Time Priority.

# Order Book – Single Venue



## Order Book - Single Venue

Quote: Bid - Ask  
\$30.00 - \$30.10  
\$30.00 x 800 - \$30.10 x 300

### Buy Orders (Bids)

| <u>Time</u> | <u>Price</u> | <u>Shares</u> |
|-------------|--------------|---------------|
| 10:00 AM    | 30.00        | 100           |
| 10:05 AM    | 30.00        | 500           |
| 10:15 AM    | 30.00        | 200           |
| 10:00 AM    | 29.90        | 500           |
| 10:20 AM    | 29.90        | 100           |
| 10:10 AM    | 29.80        | 500           |
| 10:15 AM    | 29.80        | 1000          |

### Sell Orders (Asks)

| <u>Time</u> | <u>Price</u> | <u>Shares</u> |
|-------------|--------------|---------------|
| 10:00 AM    | 30.10        | 100           |
| 10:20 AM    | 30.10        | 200           |
| 10:00 AM    | 30.15        | 500           |
| 10:10 AM    | 30.15        | 100           |
| 10:20 AM    | 30.15        | 1000          |
| 10:00 AM    | 30.20        | 500           |

# Order Book – Single Venue

## Order Book

|                    | <u>Time</u> | <u>Shares</u> | <u>Price</u> |                |
|--------------------|-------------|---------------|--------------|----------------|
| Sell Orders (Asks) | 10:00 AM    | 500           | 30.20        |                |
|                    | 10:20 AM    | 1000          | 30.15        |                |
|                    | 10:10 AM    | 100           | 30.15        |                |
|                    | 10:00 AM    | 500           | 30.15        |                |
|                    | 10:05 AM    | 200           | 30.10        |                |
|                    | 10:00 AM    | 100           | 30.10        | Best Ask Price |

Spread

|                   |          |      |       |                |
|-------------------|----------|------|-------|----------------|
| Buy Orders (Bids) | 10:00 AM | 100  | 30.00 | Best Bid Price |
|                   | 10:05 AM | 500  | 30.00 |                |
|                   | 10:15 AM | 200  | 30.00 |                |
|                   | 10:00 AM | 500  | 29.90 |                |
|                   | 10:20 AM | 100  | 29.90 |                |
|                   | 10:10 AM | 500  | 29.80 |                |
|                   | 10:15 AM | 1000 | 29.80 |                |

Sell Order Book: Asks  
Collection of All Sell Orders  
Price-Time Priority

Buy Order Book: Bids  
Collection of all Buy Orders  
Price-Time Priority

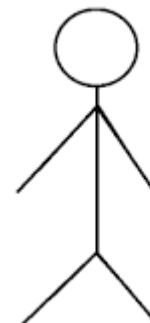
# Order Book – Buy Order

## Order Book

| Sell Orders (Asks) | Time     | Shares | Price |
|--------------------|----------|--------|-------|
|                    | 10:00 AM | 500    | 30.40 |
|                    | 10:20 AM | 1000   | 30.35 |
|                    | 10:10 AM | 100    | 30.20 |
|                    | 10:00 AM | 500    | 30.15 |
|                    | 10:05 AM | 200    | 30.10 |
|                    | 10:00 AM | 100    | 30.10 |

|                   |          |      |       |
|-------------------|----------|------|-------|
| Buy Orders (Bids) | 10:00 AM | 100  | 30.00 |
|                   | 10:05 AM | 500  | 30.00 |
|                   | 10:15 AM | 200  | 30.00 |
|                   | 10:30 AM | 100  | 30.00 |
|                   | 10:00 AM | 500  | 29.90 |
|                   | 10:20 AM | 100  | 29.90 |
|                   | 10:10 AM | 500  | 29.80 |
|                   | 10:15 AM | 1000 | 29.80 |

Buy  
Market  
Order



Buy  
Limit  
Order

Example: Buy Order

It is 10:30am. An investor wishes to buy 100 shares. She has two options:

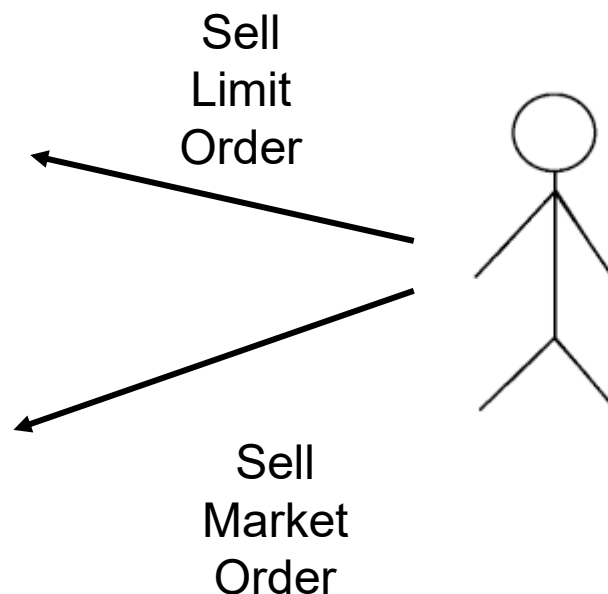
- Market Order = buy shares immediately at \$30.10 by transacting at the best ask price.
- Buy Limit Order = enter a buy limit order into the order book at the current best bid price of \$30.00. The order will not execute until all previously entered buy orders at \$30.00 are executed. There is no guarantee that this order will execute.

# Order Book – Sell Order

## Order Book

| Sell Orders (Asks) | Entry Time | Shares | Price |
|--------------------|------------|--------|-------|
|                    | 10:00 AM   | 500    | 30.40 |
|                    | 10:20 AM   | 1000   | 30.35 |
|                    | 10:10 AM   | 100    | 30.20 |
|                    | 10:00 AM   | 500    | 30.15 |
|                    | 10:30 AM   | 100    | 30.10 |
|                    | 10:05 AM   | 200    | 30.10 |
|                    | 10:00 AM   | 100    | 30.10 |

|                   |          |      |       |
|-------------------|----------|------|-------|
| Buy Orders (Bids) | 10:00 AM | 100  | 30.00 |
|                   | 10:05 AM | 500  | 30.00 |
|                   | 10:15 AM | 100  | 30.00 |
|                   | 10:00 AM | 500  | 29.90 |
|                   | 10:20 AM | 100  | 29.90 |
|                   | 10:10 AM | 500  | 29.80 |
|                   | 10:15 AM | 1000 | 29.80 |
|                   |          |      |       |



### Example: Sell Order

It is 10:30am. An investor wishes to sell 100 shares. She has two options:

- Market Order = sell shares immediately at \$30.00 by transacting at the best bid price.
- Sell Limit Order = enter a sell limit order into the order book at the current best ask price of \$30.10. The order will not execute until all previously entered sell orders at \$30.10 are executed. There is no guarantee that this order will execute.

# Order Book – Multiple Venues

## Order Book - Multiple Venues

**NBBO:** National Best Bid - National Best Ask (Offer)

\$30.02 - \$30.06

\$30.02 x 200 - \$30.06 x 100

### Venue A

**Quote:** Bid - Ask

\$30.00 - \$30.10

\$30.00 x 800 - \$30.10 x 300

| Buy Orders (Bids) |        | Sell Orders (Asks) |        |
|-------------------|--------|--------------------|--------|
| Price             | Shares | Price              | Shares |
| 30.00             | 100    | 30.10              | 100    |
| 30.00             | 500    | 30.10              | 200    |
| 30.00             | 200    | 30.15              | 500    |
| 29.90             | 500    | 30.15              | 100    |
| 29.90             | 100    | 30.15              | 1000   |
| 29.80             | 500    | 30.20              | 500    |
| 29.80             | 1000   |                    |        |

### Venue B

**Quote:** Bid - Ask

\$30.02 - \$30.15

\$30.02 x 200 - \$30.15 x 600

| Buy Orders (Bids) |        | Sell Orders (Asks) |        |
|-------------------|--------|--------------------|--------|
| Price             | Shares | Price              | Shares |
| 30.02             | 200    | 30.15              | 500    |
| 30.00             | 100    | 30.15              | 100    |
| 30.00             | 500    | 30.20              | 500    |
| 29.90             | 1000   | 30.20              | 1000   |
|                   |        | 30.30              | 100    |
|                   |        | 30.30              | 200    |

### Venue C

**Quote:** Bid - Ask

\$29.90 - \$30.06

\$29.90 x 1600 - \$30.06 x 100

| Buy Orders (Bids) |        | Sell Orders (Asks) |        |
|-------------------|--------|--------------------|--------|
| Price             | Shares | Price              | Shares |
| 29.90             | 500    | 30.06              | 100    |
| 29.90             | 100    | 30.10              | 300    |
| 29.90             | 1000   | 30.10              | 200    |
| 29.80             | 500    | 30.20              | 500    |
| 29.90             | 100    | 30.20              | 1000   |
|                   |        | 30.20              | 500    |

# Smart Order Routing

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# Smart Order Routing – Buy Order

## Market Order Book

### Venue A

| <u>Queue</u> | <u>Price</u> | <u>Shares</u> | <u>Cumul.<br/>Shares</u> |
|--------------|--------------|---------------|--------------------------|
| 1            | 30.00        | 100           | 100                      |
| 2            | 30.00        | 500           | 600                      |
| 3            | 30.00        | 1000          | 1600                     |
| 4            | 30.00        | 200           | 1800                     |
| 5            | 30.00        | 100           | 1900                     |
| 6            | 30.00        | 500           | 2400                     |
| 7            | 30.00        | 200           | 2600                     |

### Venue B

| <u>Queue</u> | <u>Price</u> | <u>Shares</u> | <u>Cumul.<br/>Shares</u> |
|--------------|--------------|---------------|--------------------------|
| 1            | 30.00        | 100           | 100                      |
| 2            | 30.00        | 500           | 600                      |
| 3            | 30.00        | 200           | 800                      |

### Venue C

| <u>Queue</u> | <u>Price</u> | <u>Shares</u> | <u>Cumul.<br/>Shares</u> |
|--------------|--------------|---------------|--------------------------|
| 1            | 30.00        | 100           | 100                      |
| 2            | 30.00        | 100           | 200                      |
| 3            | 30.00        | 500           | 700                      |
| 4            | 30.00        | 200           | 900                      |
| 5            | 30.00        | 500           | 1400                     |

### Question

- The investor wishes to enter a Buy Limit order to buy 100 shares at a price of \$30.00.
- Which venue should the investor enter the order: Venue A, Venue B, or Venue C?

# Smart Order Routing

## Market Order Book: Equal Volume at all Venues

| Venue A |       |        |               | Venue B |       |        |               | Venue C |       |        |               |
|---------|-------|--------|---------------|---------|-------|--------|---------------|---------|-------|--------|---------------|
| Queue   | Price | Shares | Cumul. Shares | Queue   | Price | Shares | Cumul. Shares | Queue   | Price | Shares | Cumul. Shares |
| 1       | 30.00 | 100    | 100           | 1       | 30.00 | 100    | 100           | 1       | 30.00 | 100    | 100           |
| 2       | 30.00 | 500    | 600           | 2       | 30.00 | 500    | 600           | 2       | 30.00 | 100    | 200           |
| 3       | 30.00 | 1000   | 1600          | 3       | 30.00 | 200    | 800           | 3       | 30.00 | 500    | 700           |
| 4       | 30.00 | 200    | 1800          | 4       | 30.00 | 100    | 900           | 4       | 30.00 | 200    | 900           |
| 5       | 30.00 | 100    | 1900          |         |       |        |               | 5       | 30.00 | 500    | 1400          |
| 6       | 30.00 | 500    | 2400          |         |       |        |               | 6       | 30.00 | 100    | 1500          |
| 7       | 30.00 | 200    | 2600          |         |       |        |               |         |       |        |               |
| 8       | 30.00 | 100    | 2700          |         |       |        |               |         |       |        |               |

- If there is equal volume traded at each venue, the investor should enter the buy limit order to Venue B because this venue has the fewest cumulative shares in the queue entered at \$30.00.

# Smart Order Routing

## Market Order Book: Venue A trades 10x more volume than B and C

| Venue A |       |        |               | Venue B |       |        |               | Venue C |       |        |               |
|---------|-------|--------|---------------|---------|-------|--------|---------------|---------|-------|--------|---------------|
| Queue   | Price | Shares | Cumul. Shares | Queue   | Price | Shares | Cumul. Shares | Queue   | Price | Shares | Cumul. Shares |
| 1       | 30.00 | 100    | 100           | 1       | 30.00 | 100    | 100           | 1       | 30.00 | 100    | 100           |
| 2       | 30.00 | 500    | 600           | 2       | 30.00 | 500    | 600           | 2       | 30.00 | 100    | 200           |
| 3       | 30.00 | 1000   | 1600          | 3       | 30.00 | 200    | 800           | 3       | 30.00 | 500    | 700           |
| 4       | 30.00 | 200    | 1800          | 4       | 30.00 | 100    | 900           | 4       | 30.00 | 200    | 900           |
| 5       | 30.00 | 100    | 1900          |         |       |        |               | 5       | 30.00 | 500    | 1400          |
| 6       | 30.00 | 500    | 2400          |         |       |        |               | 6       | 30.00 | 100    | 1500          |
| 7       | 30.00 | 200    | 2600          |         |       |        |               |         |       |        |               |
| 8       | 30.00 | 100    | 2700          |         |       |        |               |         |       |        |               |

- If Venue A trades 10x the volume as Venue B and Venue C, the investor would be best served if they entered the limit order to venue A.
- Even though Venue A has the most cumulative shares that must execute before the investor will have their shares transacted, these shares are expected to transact quicker due to the much higher volume in venue A.
- Investors must consider the expected time until they are next in que, which is dependent upon the cumulative shares already entered into the order book and the trading volume at each volume.

# Trade Venue - Revenue Model

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# Venue Revenue Models (Rebates / Fees)



- **Commission Based:**
  - Both parties pay a “**fee**” for transacting in the venue.
  - This is a common revenue model with dark pools.
- **Maker-Taker Model (“Rebate”)**
  - The investor posting liquidity (Limit Order) will receive a “**rebate**” from the trading venue. E.g., the investor who posts a limit order on the venue will receive a rebate if a trade occurs.
  - The investor taking liquidity (Market Order) is charged a “**fee**” by the trading venue. E.g., investors submitting a market order to the venue are charged a fee.
  - The difference between the Fee and Rebate is the profit for the Exchange/Venue.
- **Inverted Maker-Taker Model**
  - The investor posting liquidity (Limit Order) is charged a fee.
  - The investor taking liquidity (Market Order) receives a rebate.
  - This is the opposite of the maker-taker revenue model.

# Venue Revenue Models (Rebates / Fees)

| Revenue Model        | Maker<br>(Post Limit Order) | Taker<br>(Trade with Market Order) |
|----------------------|-----------------------------|------------------------------------|
| Maker-Taker          | Receives Rebate             | Pays Fee                           |
| Inverted Maker-Taker | Pays Fee                    | Received Rebate                    |
| Commission           | Pays Fee                    | Pays Fee                           |